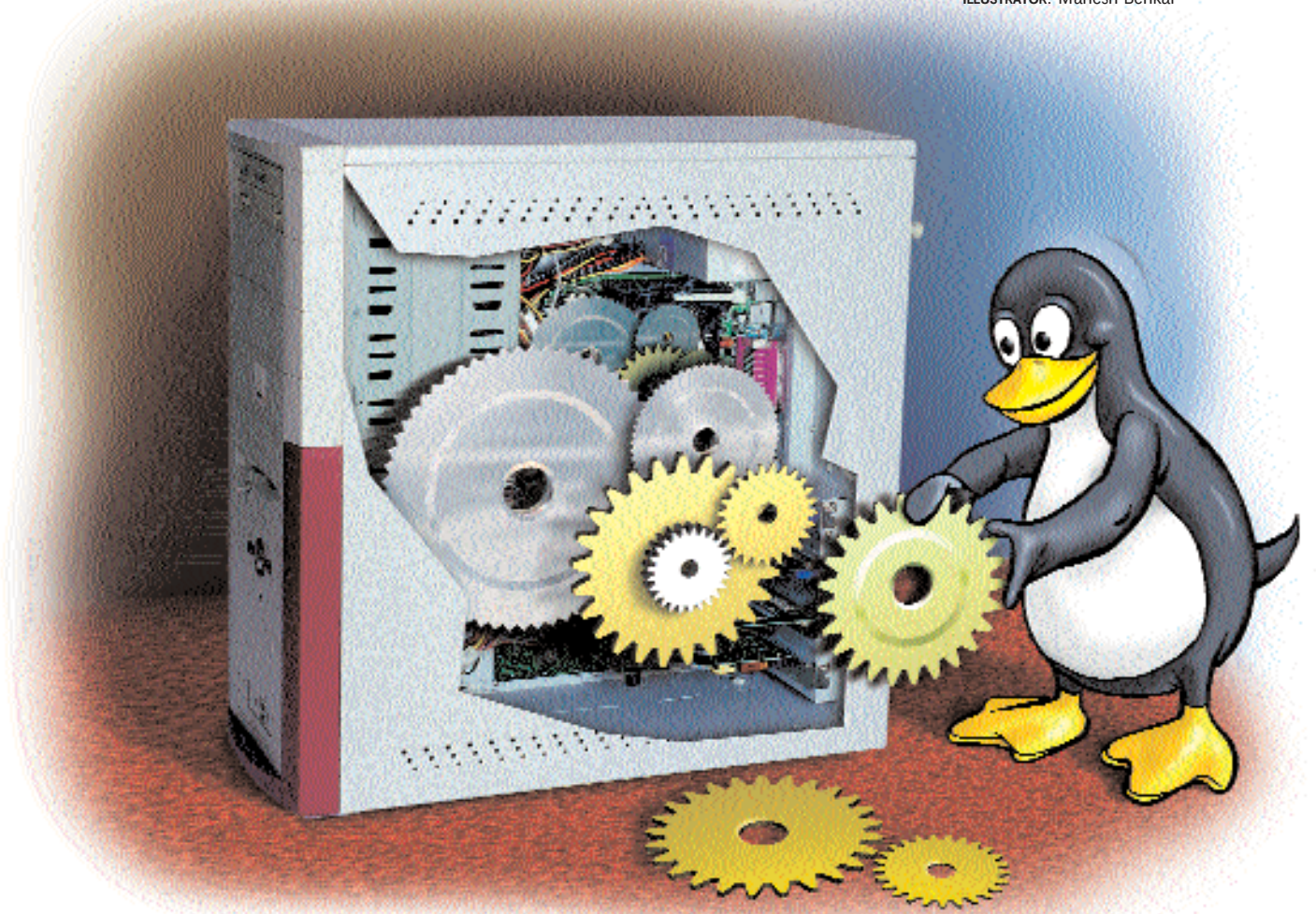




Tweaking the kernel

One of the best things about Linux is that you can chop and change anything and everything you want. the kernel—the heart of the OS—is no different, and we'll show you how to go about doing just that

Any Linux system is functionally organized at three levels: the kernel, which schedules tasks and manages storage; the shell, which connects and interprets users' commands, calls programs from memory, and executes them; and the tools and applications that offer additional functionality to the operating system.

One of the unique features of Linux is that the kernel source code is available for download, so users can compile the kernel to suit their requirements.

In this workshop we will compile and install a new version of the Linux kernel—we will keep all the other applications the same, and change just the core of the OS.

Why would anyone in his or her right mind decide to recompile the Linux kernel? There are several compelling reasons.

Imagine you've bought a brand new digital camera and it's not readily supported by your existing kernel. The vendor doesn't provide any loadable drivers, but there is support for the camera in a newer version of the kernel. If this happens, you'll have to upgrade your kernel.

Another scenario: you use a Linux desktop, and want to access NTFS partitions on Windows XP. You don't want to rely on the experimental stuff that came with your older kernel. The only alternative you have is to get hold of a new kernel with better support for NTFS, and compile it on your system.

In fact, the flexibility provided by the Linux kernel, in terms of recompilation, is one of the reasons for Linux being so widely adaptable to almost every imaginable platform and architecture.

One of the most detailed documentations available about the Linux kernel is right there in one of the kernel directories; you'll find a goldmine of information in the `/usr/src/linuxkernelversion/Documentation` directory. This directory contains individual text files for each and every file and module that is even remotely related to the kernel. While the majority of the stuff in these text files is quite techie, there's enough for the novice to get a fairly decent idea of what happens where in the kernel. But there is a small problem—this documentation is not one of those cute little html or chm files we are so used to seeing on Windows.

Installing the new kernel

To begin with, let's figure out what has and what hasn't changed in the Linux kernel. Earlier, the Linux kernel was a